DARSHAN MATHAPATI

Embedded Engineer

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CAREER OBJECTIVE

Electronics and Communication Engineering student with hands-on experience in embedded systems and IoT using ESP32, Raspberry Pi, and Arduino UNO. Skilled in sensor interfacing, cloud-connected monitoring, and hardware prototyping. Seeking an embedded engineering opportunity to build practical, real-world solutions while continuing to learn.

EDUCATION

East Point College of Engineering and Technology — **2023 – 2027**
B.E. in Electronics and Communication Engineering – 7.8 CGPA — *Bangalore, Karnataka*

ICS Mahesh PU Science College — **2021 – 2023**
Karnataka State Pre-University Board – 87.66% — *Dharwad, Karnataka*

Shri Sharada Vidyaniketan School — **2020 – 2021**
Karnataka Secondary Education Examination Board – 87.36% — *Vijayapura, Karnataka*

TECHNICAL SKILLS

Programming: C/C++ (Arduino), Python (Raspberry Pi) – basics

Microcontrollers & Boards: ESP32, Arduino UNO, Raspberry Pi 4B

Protocols: I2C, SPI, UART, Bluetooth (HC-05), Wi-Fi

Sensors & Modules: IR, Ultrasonic (HC-SR04), Gas, Fire, TDS, pH, Turbidity, Temperature, SIM800L (GSM), L298N Motor Driver, Servo Motors

Software & Tools: Arduino IDE, Thonny, Blynk IoT, Cadence, Overleaf (LaTeX)

ECE Tools: Proteus, Tinkercad, Wokwi, MATLAB, Raspberry Pi OS [Linux Based OS], ChatGPT, Claude AI

HANDS-ON EXPERIENCE

- **Raspberry Pi 4B:** Camera interfacing, OCR-based text recognition, Excel data storage, voice alerts, and communication with ESP32.
- **ESP32:** IoT monitoring with Blynk Cloud, Wi-Fi/Bluetooth communication, mobile and email notifications, servo control.
- **Arduino UNO:** Sensor interfacing, motor control using L298N, Bluetooth-controlled robotics.
- **Hardware:** Breadboard prototyping, soldering, multimeter-based testing and debugging, power regulation using buck converters.

PROJECTS

Sparsh-Vaani: Medicine Label Reader for Differently-Abled Patients — *Raspberry Pi, ESP32, OCR*

– Built an assistive system that reads medicine labels using a camera and OCR, stores details in Excel, and controls a six-compartment pill dispenser through ESP32, with voice alerts, reminders, and IoT notifications.

Smart PillCare: IoT Medicine Dispenser for Elderly — *ESP32, Blynk IoT*

– Developed an automatic dispenser with pre-alert reminders, pill dispensing, remaining-tablet count, Blynk app notifications, and email alerts to caregivers for missed doses; documented in an IEEE-format paper.

AquaVision: Smart Water Quality Monitoring — *ESP32, Blynk Cloud*

– Built an IoT system to monitor TDS, pH, turbidity, and temperature, measure water level using HC-SR04, and send mobile notifications and buzzer alerts.

Smart Agriculture System for Crop Recommendation Using IoT — *ESP32, IoT Sensors, OLED*

– Developed a final-year IoT system that monitors field conditions through sensors, recommends suitable crops to farmers, and shows results on an OLED display; documented in a VTU project report.

Disaster Management Robot — *Arduino UNO, ESP32*

– Designed a rescue robot with fire and gas detection, obstacle avoidance, Bluetooth control, IoT monitoring, and an automatic medicine dispenser; won 1st prize in an Embedded Systems Hackathon.

Bluetooth-Controlled RC Car — *Arduino UNO, HC-05, L298N*

– Built a four-motor car controlled from a smartphone app (forward, backward, left, right, stop) using an HC-05 module and L298N motor driver.

Personal Portfolio Website — *HTML, CSS, JavaScript*

– Designed and coded a responsive portfolio website from scratch, hosted online, showcasing projects, skills, leadership roles, and contact details.

LEADERSHIP & ACTIVITIES

Class Representative (CR) **2024 – Present**

Dept. of ECE, East Point College of Engineering and Technology

– Coordinate communication between students and faculty and help resolve student concerns.

IEEE Student Member

IEEE Student Branch

– Participated in IEEE Day celebrations, technical workshops, and seminars, and supported technical events.

ACHIEVEMENTS

- Participated in a National Level Hackathon (Smart Health Care and Personal Safety domain).
- 1st Prize Winner in an Embedded Systems Hackathon.
- 1st Prize in the Digital Electronics Quiz Competition by Maven Silicon during the VLSI Workshop.
- Attended technical workshops and visited exhibitions at KTPO, Whitefield, Bangalore.

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

Place: Bangalore **Darshan Mathapati**